



Circuit Breaker

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Difficulty Level : Medium

Circuit Breaker

It is 2:47 AM. The on-call engineer gets paged: response times are spiking, users are seeing errors, and the dashboard is a sea of red. At the heart of modern distributed systems lies a safety mechanism called the **circuit breaker**: when a service fails too many times within a short window, its circuit “opens” and all further requests are rejected immediately, giving it time to recover. After a cooldown period the circuit “closes” and traffic resumes normally.

You have been handed the event logs of a live production system. Replay those logs and answer the SRE team’s queries: at a given point in time, is a given service available?

Task: Given a sequence of timestamped service events and a list of queries, output for each query whether the target service’s circuit is `OPEN` or `CLOSED` at that moment.

Input

```
nbservices = N, window = W, threshold = X, cooldown = C
nb_events
events = T S R
nb_queries
queries = T S
```

- **Line 1:** Four integers: number of services N , failure window W (seconds), failure threshold X , and cooldown duration C (seconds).
- **Line 2:** Number of events
- **Events lines:** Each event is a triple $T S R$: timestamp, service ID, and result (0 = failure, 1 = success). Events are sorted by T .
- **Line 4:** Number of queries
- **Queries lines:** Each query is a pair $T S$: timestamp and service ID.

Output

For each query print `OPEN` if the service’s circuit is tripped at time T , or `CLOSED` if it is available. A circuit opens the moment a service accumulates X failures within any sliding window of W seconds. It stays open for C seconds, then closes. Events that arrive while a circuit is open are ignored and do not count toward the failure window.

Constraints

- $1 \leq N \leq 100$
- $1 \leq E, Q \leq 10^5$
- $0 \leq T \leq 10^9$
- $1 \leq W, C \leq 10^6$
- $1 \leq X \leq 10^4$
- Time limit: **2 seconds**
- Memory limit: **256 MB**

Example Input

N=2, W=10, X=3, C=15

events = [(0, 0, 0), (3, 0, 0), (7, 0, 0), (10, 0, 0), (12, 0, 1), (20, 0, 0), (25, 1, 0),]

queries = [(8, 0), (9, 0), (22, 0), (23, 0), (25, 1),]

Example Output

OPEN, OPEN, CLOSED, CLOSED, CLOSED,

Explanation

Service 0 receives failures at $t = 0$, $t = 3$, and $t = 7$. All three fall within the window $[0, 10]$, hitting the threshold $X = 3$, so the circuit opens at $t = 7$ and stays open until $t = 7 + 15 = 22$. Events at $t = 10$, $t = 12$, and $t = 20$ arrive while the circuit is open and are ignored. At $t = 22$ the circuit closes. Service 1 only has one failure at $t = 25$, well below the threshold.